

Chapter 1, Problem 31P

Problem

In a household, a business is run for an average of 6 h/day. The total power consumed by the computer and its printer is 230 W. In addition, a 75-W light runs during the same 6 h. If their utility charges 11.75 cents per kWh, how much do the owners pay every 30 days?

Step-by-step solution

Step 1 of 2

In a household, a computer and its printer consume **230 W** and a light consumes **75 W**. Both of them run 6 hours a day.

Calculate the total power consumed.

$$\begin{aligned} p &= 230 + 75 \\ &= 305 \text{ W} \end{aligned}$$

Calculate the total energy consumed by computer printer and light

$$\begin{aligned} w &= pt \\ &= (305 \text{ W})(6 \text{ h}) \\ &= 1.83 \text{ kWh} \end{aligned}$$

Step 2 of 2

The utility charges **11.75 cents** per kWh.

Calculate the total utility charges for 30 days.

$$\begin{aligned} \text{utility charges} &= (11.75 \text{ cents/kWh})(1.83 \text{ kWh})(30) \\ &= 645 \text{ cents} \end{aligned}$$

Therefore, the charges owners pay every 30 days is **645 cents**.